CURRENT ENGINE WORKFLOW

Build a Fully Animated Character

A start-to-finish setup for idle, walking, running, jumping, falling and landing using the Clip Editor, Graph Editor and Character Editor.

Clip assets > Animation graph > Character asset > Play mode

What you will create

A reusable character asset with a skeletal mesh, capsule collision, player movement and a graph-driven animation state machine. Ground movement blends continuously through `Idle -> Walk -> Run`. Pressing Space switches to `Jump`, then `Fall`, then `Land` before returning to locomotion.

Asset	Purpose	Recommended folder
`.3dgclip`	Names and configures one animation take.	`Content/Assets/AnimationClips`
`.3dggraph`	Owns clips, parameters, states, blend spaces and transitions.	`Content/Assets/Animations`
`.3dgcharacter`	Combines the model, collider, movement and graph.	`Content/Assets/Characters`

IMPORTANT ARCHITECTURE
Animation is authored in the Graph Editor. The Character Editor references the saved graph; do not rebuild the same locomotion as legacy inline animation sources.

Before starting

You need one skeletal character model and animation files that share the same skeleton and bone names. A Mixamo-style separate-FBX workflow is supported.

Required	Loop?	Root motion
Idle	Yes	Usually unchanged
Walk	Yes	Strip for controller-driven movement
Run	Yes	Strip for controller-driven movement
Jump start	No	Strip
Fall loop	Yes	Strip
Land	No	Strip

1. Import and create animation clips

Keep every source inside the project's Content folder so it appears in searchable editor pickers and is available to packaged builds.

1

Import the skeletal model and FBX animations

Use the Assets panel Import control, then Refresh. A practical layout is `Content/Assets/Characters` for the mesh and `Content/Assets/Animations` for the source FBX files.

2

Open the Clip Editor

Open `Panels -> Clip Editor`. Create one `.3dgclip` for every required animation. A saved clip can also be reopened by double-clicking it in Assets.

3

Configure each clip

Set Name to the gameplay name, choose Source File, then select the source take. FBX files often call the internal take `Unreal Take`; that is valid because the clip asset keeps the friendly name separately.

4

Set playback behavior

Enable Loop for Idle, Walk, Run and Fall. Disable Loop for Jump and Land. Enable Strip Root Motion on controller-driven locomotion and air clips when the source translates the root.

5

Save with matching filenames

Save as `Idle.3dgclip`, `Walk.3dgclip`, `Run.3dgclip`, `Jump.3dgclip`, `Fall.3dgclip` and `Land.3dgclip`.

Clip asset	Name	Source take	Loop	Strip root
Idle.3dgclip	Idle	Selected FBX take	Yes	Optional
Walk.3dgclip	Walk	Selected FBX take	Yes	Yes
Run.3dgclip	Run	Selected FBX take	Yes	Yes
Jump.3dgclip	Jump	Selected FBX take	No	Yes
Fall.3dgclip	Fall	Selected FBX take	Yes	Yes
Land.3dgclip	Land	Selected FBX take	No	Yes

ROOT MOTION
The player controller moves the capsule. If Walk or Run also moves the animation root, the mesh double-moves and slides. Strip Root Motion keeps the animation in place.

Clip preview checks

Use a Preview Mesh when the animation FBX has no skin. Confirm the correct take plays, the pose matches the skeleton, and the character remains centered when root motion is stripped.

2. Create the animation graph

6

Open the Graph Editor and select a Preview Rig

Create a new graph, set its name, choose the character skeletal model as Preview Rig, and save it as `PlayerLocomotion.3dggraph`.

7

Add all six clip assets

Under Clips, add Idle, Walk, Run, Jump, Fall and Land. The graph displays their friendly asset names while retaining the internal source take.

8

Declare runtime parameters

Add the parameters below with the exact spelling and type shown. Names are case-sensitive.

Parameter	Type	Runtime meaning
`Speed`	Float	Horizontal movement speed.
`IsGrounded`	Bool	True while the capsule has ground contact.
`IsFalling`	Bool	True while airborne with negative vertical velocity.
`VerticalSpeed`	Float	Positive while rising and negative while falling.

9

Create the grounded blend space

Click `Create 1D Locomotion`. Keep the state name `Locomotion`, choose 1D Blend Space, set X Parameter to `Speed`, and enable Synchronize Samples.

Recommended 1D blend samples

Clip	Speed sample	Purpose
Idle	0.0	Standing still
Walk	3.0	Normal movement
Run	6.0	Sprint

MATCH MOVEMENT VALUES
The Walk and Run sample coordinates should match the Character Editor's Walk Speed and Run Speed. If movement uses 3 and 6, use 3 and 6 in the blend space.

Preview the ground blend

Drag the Graph Editor's `Speed` preview value from 0 to 6. Confirm a smooth Idle-to-Walk-to-Run blend and check foot timing at intermediate values.

3. Add jump, fall and landing states

10 Create three air states

Add Jump, Fall and Land. Assign the matching clips. Jump and Land are non-looping; Fall loops until ground contact.

State	Clip	Loop	Speed
Jump	Jump	No	1.0
Fall	Fall	Yes	1.0
Land	Land	No	1.0

Locomotion > Jump > Fall > Land > Locomotion

Recommended transitions

From -> To	Conditions	Match	Fade / Exit
Locomotion -> Jump	`IsGrounded == false` `VerticalSpeed >= 0.1`	All (AND)	Fade 0.08, Interrupt
Locomotion -> Fall	`IsGrounded == false` `VerticalSpeed < 0`	All (AND)	Fade 0.10, Interrupt
Jump -> Fall	`IsFalling == true`	All	Fade 0.10
Jump -> Land	`IsGrounded == true`	All	Fade 0.08
Fall -> Land	`IsGrounded == true`	All	Fade 0.08
Land -> Locomotion	`IsGrounded == true`	All	Fade 0.12, Exit 0.85

WHY TWO CONDITIONS?
Locomotion -> Jump uses both ground state and upward velocity. Locomotion -> Fall uses ground state and downward velocity. This separates a deliberate jump from walking off a ledge.

How to add a multi-condition transition

11 Create the transition and choose From/To

In Transitions, select the source and destination states.

12 Configure Condition 1

Select the first parameter, comparison and value. For a Bool, choose `is true` or `is false`.

13 Add Condition 2

Click `Add Condition`, select the second parameter and comparison, then set Match to `All conditions (AND)`. Use OR only when either condition should independently allow the transition.

4. Assemble the character asset

14 Open the Character Editor

Create a new character and save it under `Content/Assets/Characters`.

15 Assign the skeletal model and material

Choose the same model used as the graph preview rig. Optionally select a saved material.

16 Correct model orientation

If the model lies face-down, apply the Z-up stand-up correction. Adjust render-only yaw until the character faces the preview camera and aligns with the editor's forward arrow.

17 Configure the capsule collider

Enable collision and use Capsule. Fit Radius around the torso and Half Height from feet to head. Keep the collider upright; model orientation is render-only.

18 Enable and tune the Player Controller

Use Walk Speed `3.0`, Run Speed `6.0`, Jump Speed `5.0` as a starting point. Tune capsule radius, capsule height, slope and step height for the level.

19 Assign the animation graph

In the Animation Graph section select `PlayerLocomotion.3dggraph`. The summary should report the expected states and transitions.

20 Save and add to the scene

Save the character asset, then use Add to Scene or drag it from Assets. Place it above a floor that has collision.

Character setting	Starting value	Must agree with
Walk Speed	3.0	Walk blend sample = 3.0
Run Speed	6.0	Run blend sample = 6.0
Jump Speed	5.0	Jump height and animation timing
Capsule	Fit the body	Visible collider boundary
Animation Graph	PlayerLocomotion.3dggraph	Graph clip skeleton

SKELETON COMPATIBILITY
If a clip produces a T-pose or distorted limbs, its bone names or hierarchy do not match the character model. Re-export the animation using the same rig.

5. Test the complete character

Enter Play mode and validate in this order. Testing one behavior at a time makes transition errors easy to isolate.

Test	Input	Expected animation
Idle	No movement	Idle sample at Speed 0
Walk	W/A/S/D	Blend to Walk
Run	Movement + Shift	Blend to Run
Jump rise	Space	Locomotion -> Jump
Apex/descent	Continue airborne	Jump -> Fall
Landing	Touch ground	Fall -> Land
Recovery	Land clip reaches exit time	Land -> Locomotion
Ledge fall	Walk off an edge	Locomotion -> Fall, without Jump

Runtime-driven parameters

The editor Play mode and packaged player update the graph automatically. `Speed` comes from movement input and walk/run speed; `VerticalSpeed` comes from capsule Y velocity; `IsGrounded` and `IsFalling` come from the character controller. No gameplay script is required for basic locomotion and jumping.

Troubleshooting

Symptom	Fix
All clips appear as Unreal Take	Re-add the saved .3dgclip assets to the graph. Friendly clip asset names are separate from the internal FBX take.
Character stays in bind/T-pose	Confirm the graph is selected in the Character Editor and all clips use the same skeleton.
Idle plays but Walk/Run do not	Check exact Speed parameter spelling and match blend sample values to Walk/Run Speed.
Physical jump works but animation stays grounded	Declare IsGrounded and VerticalSpeed with exact names and verify Locomotion -> Jump conditions.
Jump plays when walking off a ledge	Use AND conditions: upward VerticalSpeed for Jump, negative VerticalSpeed for Fall.
Stuck in Fall or Land	Add IsGrounded == true into Land and set Land -> Locomotion Exit Time around 0.85.
Feet slide or mesh drifts	Enable Strip Root Motion on Walk, Run and relevant air clips.
Character faces backward or lies down	Correct the Character Editor model orientation; do not rotate the capsule to compensate.
Transition feels abrupt	Increase Fade gradually, usually 0.08 to 0.18 seconds.

Final setup checklist

Done	Check
[ ]	Model and animation FBX files share the same skeleton and bone names.
[ ]	Idle, Walk, Run, Jump, Fall and Land are saved as named .3dgclip assets.
[ ]	Walk and Run have Strip Root Motion enabled when required.
[ ]	The graph contains all six clip assets and a valid Preview Rig.
[ ]	Speed, IsGrounded, IsFalling and VerticalSpeed are declared with exact spelling.
[ ]	Locomotion is a synchronized 1D Speed blend space.
[ ]	Walk/Run samples match Character Editor Walk/Run speeds.
[ ]	Jump and Land do not loop; Fall does loop.
[ ]	Jump and ledge-fall transitions use separate multi-condition rules.
[ ]	The Character Editor references the saved .3dggraph.
[ ]	Capsule fits the body and the model faces the forward arrow.
[ ]	Idle, walking, running, jump, fall, land and ledge fall all pass in Play.

A RELIABLE BASELINE
Once this graph works, extend it with start/stop states, 2D directional locomotion, crouch, attacks, layered upper-body actions and animation events. Keep basic movement stable before adding combat.

Reference values

Item	Baseline
Walk / Run / Jump Speed	3.0 / 6.0 / 5.0
Locomotion samples	Idle 0.0, Walk 3.0, Run 6.0
Air transition fades	0.08 to 0.12 seconds
Land exit time	0.85
Ground threshold	IsGrounded Bool
Rise / fall split	VerticalSpeed >= 0.1 / VerticalSpeed < 0

End of guide